

João Paulo Antunes de Souza

Full Stack Developer · Computer Engineer

Nationality: Brazilian

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Work model: Remote only

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ABOUT ME

Computer Engineer and Full Stack Developer with 6+ years of experience building cloud software for precision agriculture — from React/Next.js interfaces and serverless Python APIs on AWS to a self-hosted CI/CD server for embedded firmware. I own features end to end: data model, API, user interface, deployment, monitoring and cost. 1,000+ merged pull requests. Experienced with geospatial data, real-time IoT telemetry, WhatsApp integrations and AI-assisted development (Claude Code, ChatGPT).

WORK EXPERIENCE

08/2022 – Present

Full Stack Developer (Mid-level)

Zait (formerly Smart Agri / Smart Sensing) — Piracicaba, Brazil

Work across several of the company's precision agriculture products.

- Coverage path planning over field boundaries with Fields2Cover and Shapely: obstacle avoidance, AB-line routes, sampling routes and export to QGIS and interactive HTML maps.
- Multi-machine operations and real-time telemetry in the web UI (Next.js, React, TypeScript, Leaflet, MQTT, AWS IoT Core).
- Authentication and role-based access control on Amazon Cognito: user lifecycle, permission groups and server-side token validation.
- WhatsApp Business Cloud API (Meta) integration: automatic alerts routed to each machine's contacts, deduplication, and interactive command menus with confirmation.
- Designed and built a self-hosted CI/CD server (Python, systemd, Docker) that replaced AWS CodeBuild for building and publishing embedded firmware (ESP-IDF, PlatformIO, STM32), with reproducible builds and Discord notifications; plus automated backend deploys with rollback.
- Reduced AWS cost through measurement (Cost Explorer, CloudWatch): Lambda right-sizing, data retention with TTL, key lookups instead of scans and S3 lifecycle rules.
- Field reports rendered in the cloud (Matplotlib, Playwright), GeoTIFF layers and report delivery over WhatsApp.
- DynamoDB single-table data modelling and infrastructure as code with AWS CDK (Python).

12/2021 – 08/2022

Full Stack Developer (Junior)

Zait (formerly Smart Agri) — Piracicaba, Brazil (hybrid)

- Map-based web interfaces in React and Leaflet, with GeoJSON and drone imagery layers.
- REST APIs in Python (FastAPI) on AWS Lambda and API Gateway with DynamoDB, S3 and Cognito; unit tests; PostgreSQL, MySQL and DynamoDB data modelling.

12/2020 – 12/2021

IT Intern

Zait (formerly Smart Agri) — Piracicaba, Brazil

- REST APIs in Python (FastAPI) and React interfaces; Docker and AWS serverless services (Lambda, API Gateway, DynamoDB, S3).
- Database modelling and unit testing.

06/2019 – 09/2020

Junior Systems Analyst

Farm Solutions — Piracicaba, Brazil

- Web development (HTML, CSS, JavaScript) with Git version control.

EDUCATION AND TRAINING

01/2018 – 12/2023

Bachelor's Degree in Computer Engineering

Federal Institute of Education, Science and Technology of São Paulo (IFSP) — Brazil

- Competitive programming training group.
- Windows application development group (C#, MySQL).

2015 – 2017

Technical Course in Electronics

Etec Prof. Armando Bayeux da Silva — Brazil

LANGUAGE SKILLS

Mother tongue: Portuguese

	Listening	Reading	Spoken production	Spoken interaction	Writing
English	B1	C1	B1	B1	B2

Levels: A1/A2 basic · B1/B2 independent · C1/C2 proficient (Common European Framework of Reference)

DIGITAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C, C#

Front end: React, Next.js, React Native, Leaflet, HTML, CSS

Back end & cloud: FastAPI, REST APIs, AWS (Lambda, API Gateway, DynamoDB, Cognito, S3, IoT Core/MQTT, EventBridge, CloudWatch, CodeBuild, ECR, Secrets Manager, Cost Explorer), AWS CDK, Docker

Data & GIS: DynamoDB single-table design, PostgreSQL, MySQL, GeoJSON, Shapely, Fields2Cover, GeoTIFF, QGIS, QField, Matplotlib

DevOps & quality: Git, GitHub, CI/CD, pytest, pre-commit, Linux, systemd

Integrations: WhatsApp Business Cloud API, Telegram Bot API, Google OAuth, webhooks

Embedded: ESP-IDF, PlatformIO, STM32 (build tooling), MicroPython, ESP32, Arduino

AI tools: AI-assisted development with Claude Code and ChatGPT; Claude API

PERSONAL PROJECTS

Event promoter app (PWA): Offline-first mobile app to log the workday in seconds: Google sign-in, PDF export, AI form filling from a photo, push reminders and automatic manager spreadsheets. In real use. (TypeScript, serverless, PostgreSQL, Claude API)

Wireless screen casting: Casts a computer screen to phones or browsers, locally or over the internet, with QR-code pairing. (Node.js, WebRTC, WebSocket, Cloudflare Workers)

Running app: Web app for runners with maps and Google sign-in, on its own API. (React, TypeScript, Leaflet, FastAPI, PostgreSQL)

Finance Telegram bot: Logs expenses through a guided chat into a cloud database. (Node.js, serverless, PostgreSQL)

Business website (freelance): Responsive site for a healthcare professional. (Next.js, Tailwind CSS)

Oh My Map: Web map with area drawing, address search and routing (Next.js, Leaflet) — oh-my-map.vercel.app